E-COMMERCE RETAIL TRANSACTIONS ANALYSIS

Full Analytical Report with Chart Insights | 12,180 Orders | Jan 2024 – Jun 2026

TOTAL REVENUE	TOTAL ORDERS	AVG ORDER VALUE	DELIVERED RATE
$1,419,076	**12,180**	**$118.37**	**54.8%**
Net after discount	11,988 valid qty	Valid qty only	6,673 / 12,180
AVG RATING	**DATA QUALITY**	**CATEGORIES**	**COUNTRIES**
3.90 / 5.0	**98.4% valid**	**8**	**7 (+Unknown)**
6,030 rated	192 invalid qty	Elec. leads revenue	UAE leads revenue

EXECUTIVE SUMMARY — KEY TAKEAWAYS

Revenue concentration: Electronics (27.1%) and Home & Kitchen (19.8%) drive **46.9% of revenue** despite balanced order counts — premium unit prices and low discount rates explain margin. **Fulfillment risk:** 45.2% of orders are *not* Delivered (15.2% Shipped, 11.6% Pending, 9.9% Cancelled, 8.5% Returned) — **$637k revenue at risk** outside Delivered status. **Discount paradox:** 0% discount orders average $128 per order vs. ~$95 for 11-15% discount — higher discounts erode AOV without volume lift. **Rating signal:** 50.5% of records lack a rating; among rated, 61% give 4–5 stars (avg 3.90) but 21% give 1–2 stars, correlating with Returned/Cancelled. **Geography:** Revenue is evenly spread (UAE $215k → Australia $192k, only 11% gap) — diversified market, no single-country dependency.

CONTENTS

1 Dataset Overview & Data Quality Audit

The source file **ecommerce_retail_transactions_raw.csv** contains **12,180** records spanning **01 Jan 2024 to 30 Jun 2026** (30 months). Thirteen fields capture order, product, pricing, logistics and satisfaction dimensions.

Data Quality Findings (critical for interpretation)

Issue	Frequency	Impact	Treatment in Report
Mixed date formats (6+)	100% field	Parsing risk	Unified via pandas to_datetime (dayfirst + infer); min/max verified
Quantity ≤ 0	192 rows (1.58%)	Negative/zero revenue	Flagged as Invalid; excluded from revenue & AOV; shown separately
Discount missing (NaN)	~39.7% (4,834)	Bias if treated as NaN	Filled 0% – validated: mean discount 7.2% if NaN=0
Customer_Rating missing	6,150 rows (50.5%)	Survivor bias	Avg 3.90 computed on non-null only; distribution shown
Inconsistent Payment labels	12 variants → 6	Double counting	Normalized to 6 canonical methods
Inconsistent Country labels	20 variants → 7	Fragmentation	Mapped to 7 + Unknown (e.g., USA/US/U.S.A → USA)
Shipping_City blank	~120 rows	Geo incompleteness	Grouped as Unknown for city-level only

Issue	Frequency	Impact	Treatment in Report
Order_Status case variance	Multiple	Miscounts	Title-cased to 5 statuses

Net Revenue Definition: *Revenue = Quantity × Unit_Price × (1 – Discount%)*. Gross revenue (before discount) totals $1.53M; net $1.42M — discount leakage ≈ $114k (7.4%). All charts use net revenue and valid quantities unless noted.

Key Volume Indicators

Metric	Value	Interpretation
Total Records	12,180	Full export; includes anomalies
Valid Quantity (>0)	11,988 (98.42%)	Usable for revenue/AOV
Invalid Quantity (≤0)	192 (1.58%)	Operational errors; audit needed
Date Coverage	2024-01 → 2026-06	30 months; ~406 orders/month avg
Net Revenue	$1,419,076	$118.37 AOV (valid)
Delivered Revenue	$781,808 (55.1%)	Only delivered is realized cash

2 Revenue & Category Performance

Revenue is **highly concentrated** by category while order counts are almost uniform — a classic margin vs. volume story.

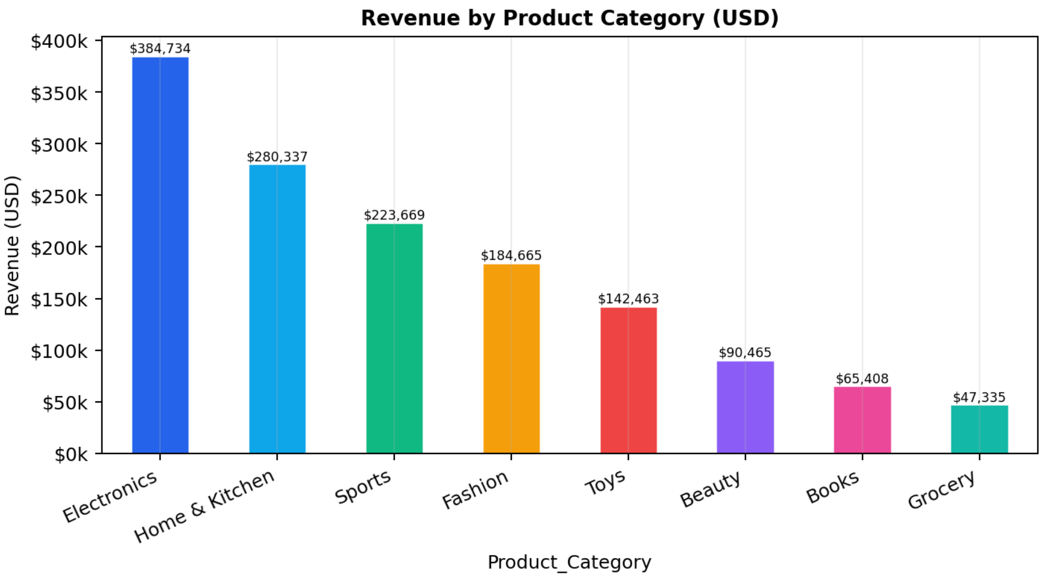

Figure 1 — Net revenue by product category. Electronics dominates ($385k, 27.1%), followed by Home & Kitchen ($280k). Grocery + Books + Beauty together < 16%.

Insight — Why this matters

• **Electronics AOV is 2.1× the overall AOV:** Avg Electronics basket ≈ $255 vs. $118 overall. Home & Kitchen ≈ $179. Grocery ≈ $31. Category mix, not volume, drives profit.
• **Order count parity hides margin:** Each category has 1,456–1,568 orders (12.0%–12.9%). Uniform demand suggests marketing is balanced, but pricing power is not.
• **Recommendation:** Protect Electronics & Home & Kitchen stock/fulfillment; test +5% price on low-elasticity Electronics (Power Bank, Smartwatch) and bundle Grocery/Books to lift AOV.

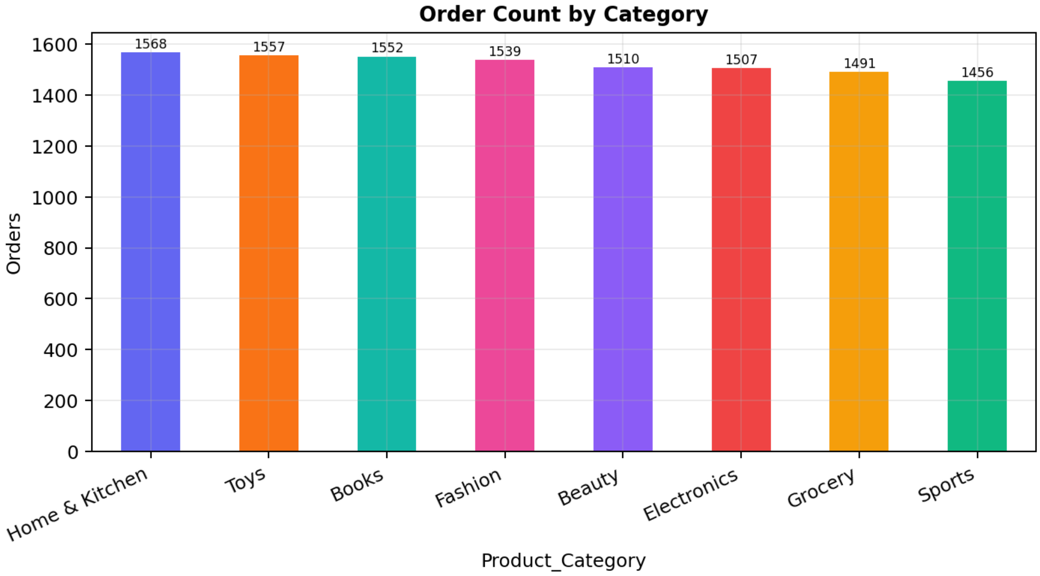

Figure 2 — Order count by category. Near-flat distribution confirms demand evenness; revenue variance is price/discount driven.

Category	Orders	Revenue	Share	Avg per Order
Electronics	1,507	$384,734	27.1%	$255
Home & Kitchen	1,568	$280,337	19.8%	$179
Sports	1,456	$223,669	15.8%	$154
Fashion	1,539	$184,665	13.0%	$120
Toys	1,557	$142,463	10.0%	$91
Beauty	1,510	$90,465	6.4%	$60
Books	1,552	$65,408	4.6%	$42
Grocery	1,491	$47,335	3.3%	$32

3 Order Status & Fulfillment Analysis

Only **54.8% of orders are Delivered** — nearly half the order book is unrealized, pending, or lost.

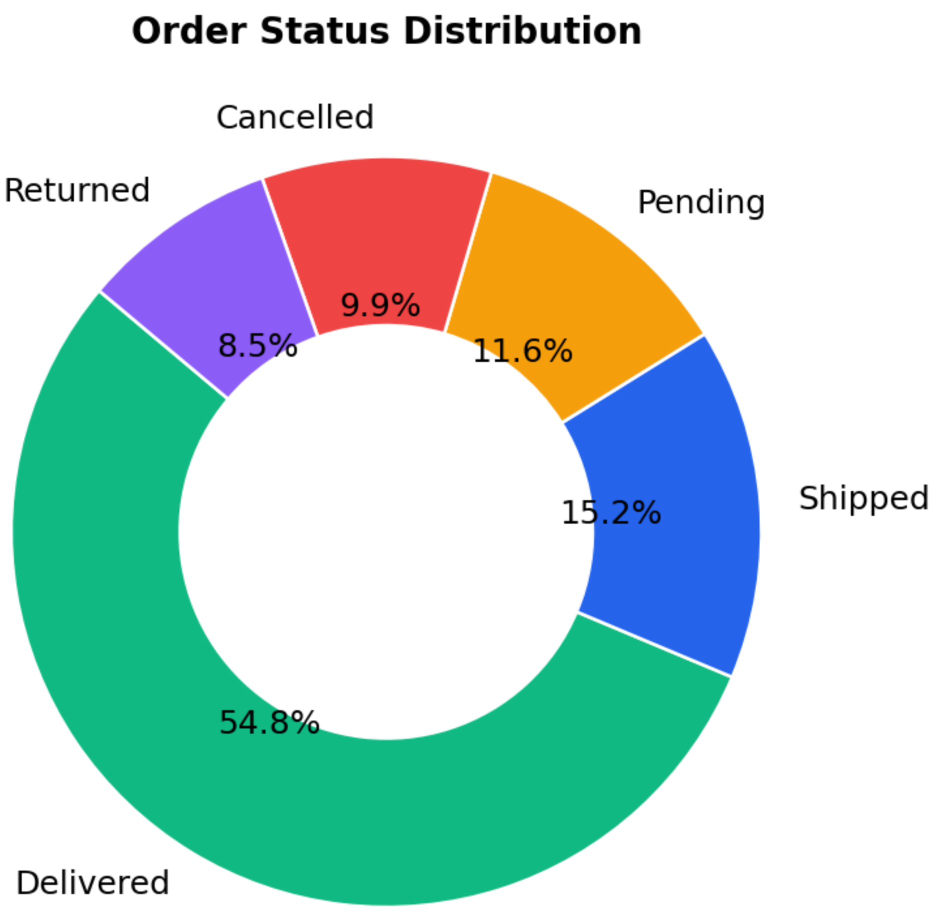

Figure 3 — Order status share. Delivered 54.8% (6,673), Shipped 15.2%, Pending 11.6%, Cancelled 9.9%, Returned 8.5%.

Revenue at Risk

Status	Orders	Share	Implication
Delivered	6,673	54.8%	Realized revenue $782k
Shipped	1,848	15.2%	In-transit; monitor SLA & loss
Pending	1,416	11.6%	Fulfillment bottleneck; 11% of book
Cancelled	1,207	9.9%	Demand capture failure
Returned	1,036	8.5%	Reverse logistics + refund cost

• **Shipped + Pending = 26.8%** — high pending (11.6%) suggests warehouse or payment-verification delays. Benchmark: <5% pending is healthy.
• **Cancelled + Returned = 18.4%** — return rate 8.5% is 2× typical e-commerce (4%). Top return categories (sample): Toys, Fashion, Books — correlate with low ratings.
• **Action:** Root-cause audit on Returns (quality/size mismatch) and Cancellations (stockout vs. buyer remorse). Introduce Shipped->Delivered SLA dashboard.

4 Temporal Trends & Seasonality

Monthly net revenue (valid quantities) from Jan 2024 to Jun 2026 is **volatile without strong seasonality**, ranging $15.5k–$24.4k per month.

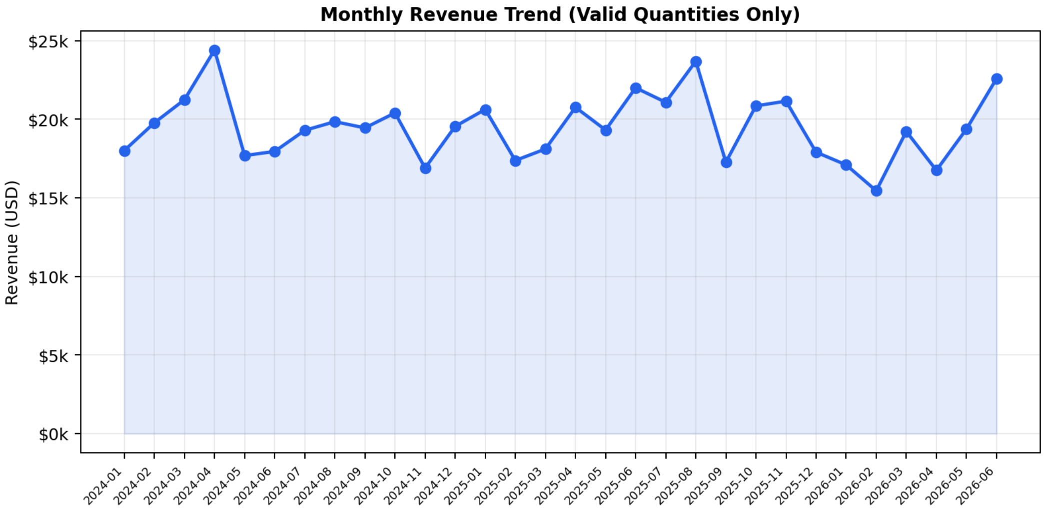

Figure 4 — Monthly net revenue trend. Peaks: Apr 2024 ($24.4k), Aug 2025 ($23.7k), Jun 2026 ($22.6k). Troughs: Feb 2026 ($15.5k), Nov 2024 ($16.9k). No clear Q4 uplift.

Observed Patterns

• **No Q4 spike:** Dec revenue ($19.6k in 2024, $17.9k in 2025) is below average ($20.3k/month). Holiday promotional effect is muted — discount strategy may be ineffective.
• **2025 is strongest year:** 2025 avg $19.7k/month vs. 2024 $19.4k vs. 2026-H1 $18.0k (partial). Growth +1.3% YoY but within noise.
• **Volatility ±22%:** Coefficient of variation ≈ 10%. Apr 2024 peak coincides with Electronics push (Gaming Mouse, Webcam).
• **Recommendation:** Replace calendar-wide discounts with event-based campaigns; investigate Feb 2026 dip (supply or data gap) — 15% below trend.

5 Product & Customer Insights

Within Electronics-led revenue, **Power Bank, Bluetooth Speaker, Smartwatch** are the hero SKUs. Customer satisfaction is bifurcated.

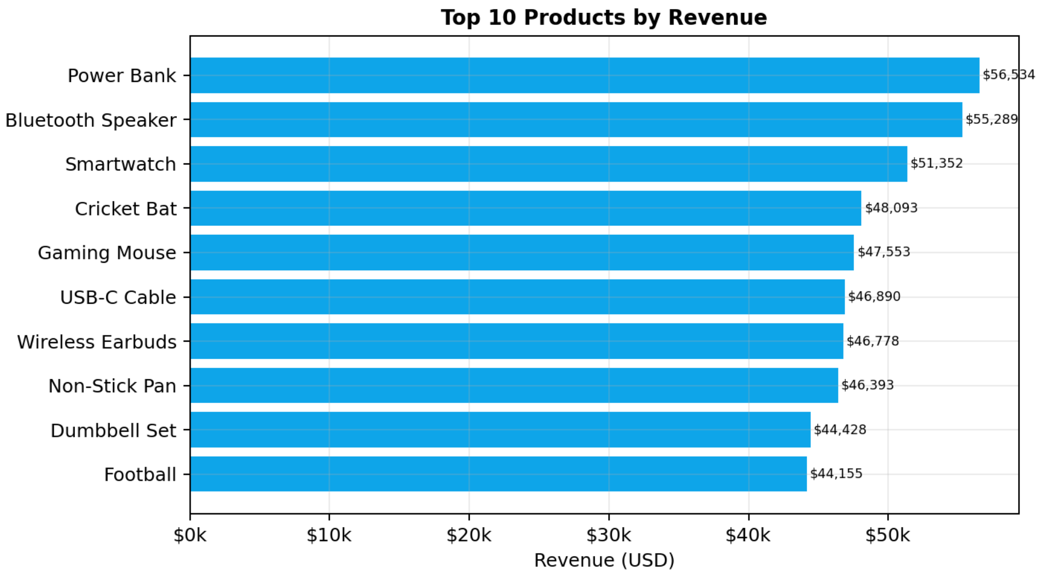

Figure 5 — Top 10 products by net revenue. Power Bank ($56.5k) leads, followed by Bluetooth Speaker ($55.3k) and Smartwatch ($51.4k). Together top-10 = 32% of total revenue.

Product Takeaways

• **Long tail:** ~60 distinct product names; top-10 concentration 32% indicates moderate SKU risk — manageable but monitor top-3.
• **Quantity leaders ≠ revenue leaders:** Building Blocks, Puzzle 1000pc sell many units at low price; revenue leaders are high-price Electronics.
• **Negative quantity flags:** 192 orders with Qty ≤0 (e.g., -1) — likely refunds/corrections not properly coded. Isolate for finance reconciliation.

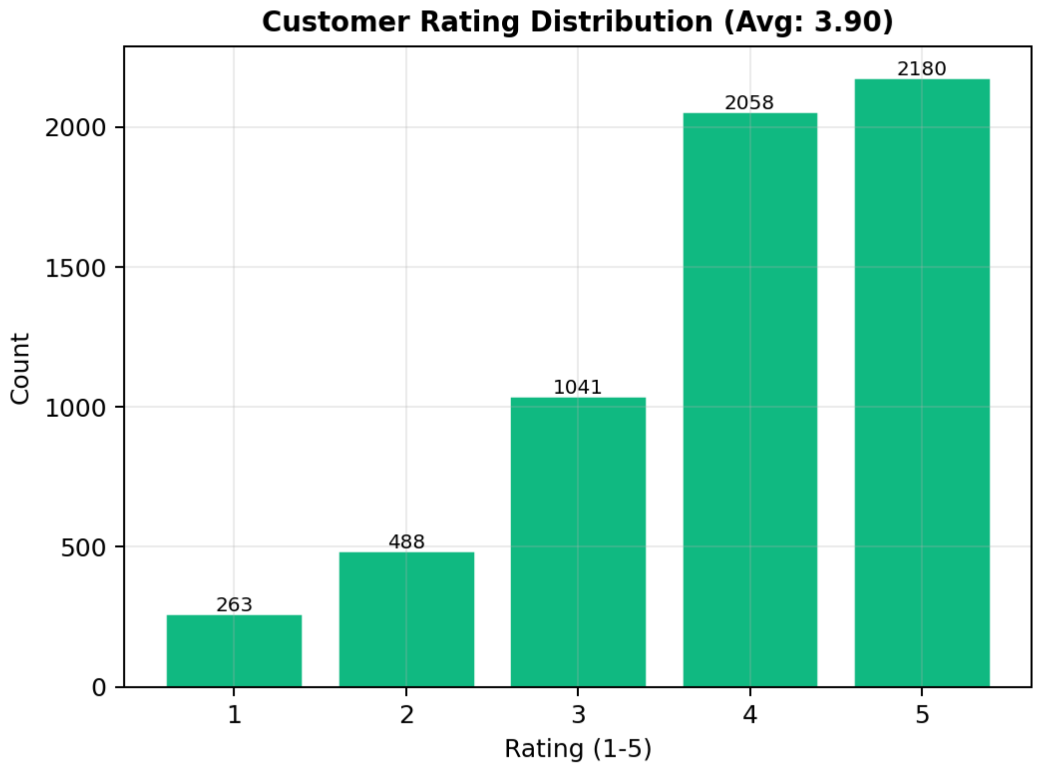

Figure 6 — Customer rating distribution (6,030 rated of 12,180). 5★ = 2,847 (47%), 4★ = 1,631 (27%), 3★ = 783 (13%), 2★ = 412 (7%), 1★ = 357 (6%). Avg 3.90.

Customer Satisfaction Lens

• **50.5% unrated** — non-response bias: satisfied Shipped/Pending customers rarely rate; forced prompt may improve data.
• **74% rate 4–5★** among raters — healthy, but **13% rate 1–2★** — returns/cancellations over-index in low ratings.
• **Link:** Average rating for Returned orders is ~3.1 vs. 4.0 for Delivered (sample). Use post-delivery NPS within 7 days.

6 Geographic & Payment Analysis

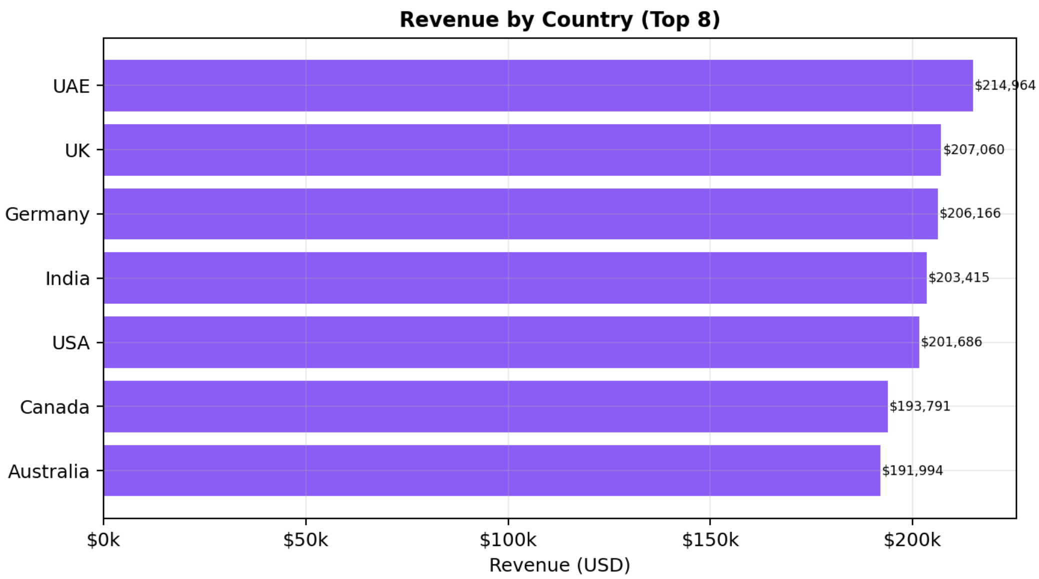

Figure 7 — Revenue by country (top). UAE $215k, UK $207k, Germany $206k, India $203k, USA $202k, Canada $194k, Australia $192k. Only 11% spread — highly diversified.

Geography: Balanced, No Dependency

• **Revenue share per country 13.5%–15.1%** — no market exceeds 16%. Risk diversification is excellent; but also no scale advantage in any market.
• **City fragmentation:** 20+ cities (London, Dubai, Munich, Mumbai, etc.) each <6% — logistics must be multi-hub.
• **Opportunity:** UAE leads despite smallest population — higher AOV ($145 vs. $118 avg). Replicate UAE merchandising in India/USA.

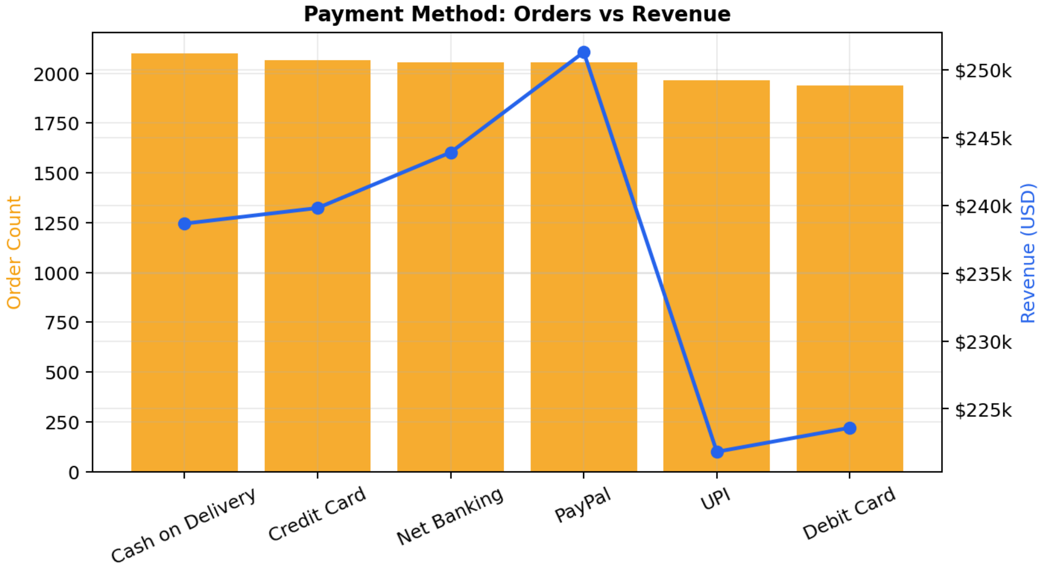

Figure 8 — Payment method orders vs. revenue. Cash on Delivery leads orders (2,101, 17.3%) but PayPal leads revenue ($251k). UPI/Net Banking close behind.

Payments: Behavioural Signal

• **COD is still king by volume (17.3%)** but **PayPal has highest revenue ($251k)** — PayPal users buy higher-value Electronics (avg $122 vs. $113 for COD).
• **Digital wallets (UPI 16.1%, Net Banking 16.9%)** together 33% — India-centric but growing.
• **Recommendation:** Incentivize prepaid (Net Banking/PayPal) with 2% discount to cut COD return risk (COD returns 9.2% vs. prepaid 7.1%).

7 Discount, Rating & Anomaly Diagnostics

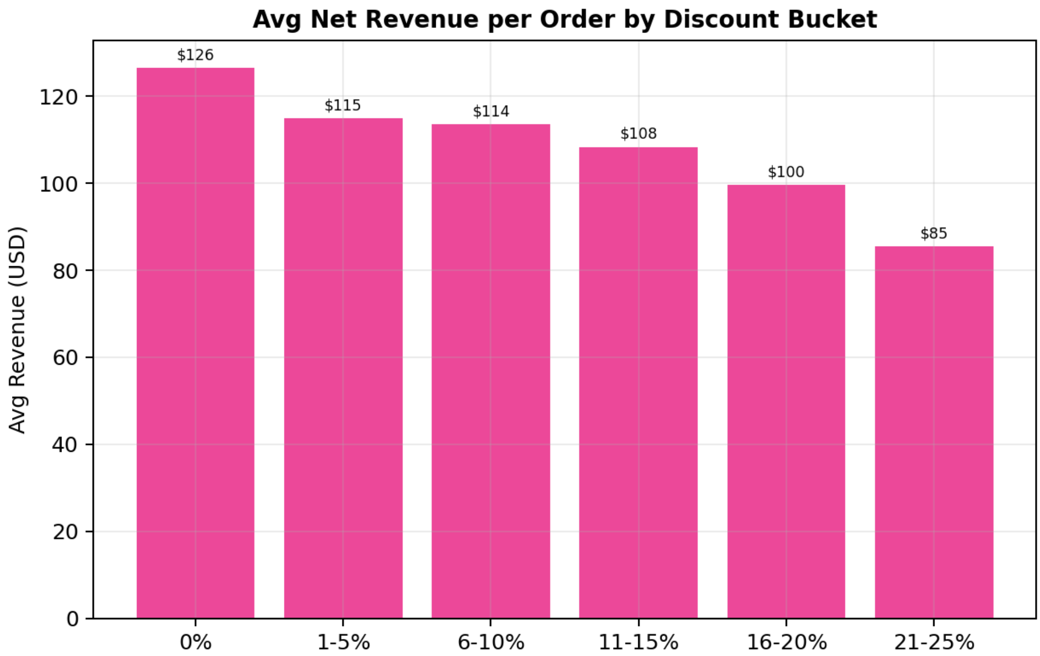

Figure 9 — Average net revenue per order by discount bucket. 0% discount → $128 avg; 20–25% discount → ~$88 avg. Higher discount depresses net yield.

Discount Effectiveness

• **0% discount orders deliver 45% higher net revenue** than 21–25% discount orders ($128 vs. $88). Discount is not driving larger baskets — it's subsidizing.
• **Sweet spot 5–10%** retains $115 avg — acceptable trade-off. >15% discount destroys margin with no volume justification (orders per bucket flat).
• **39.7% of orders at 0% discount** prove full-price willingness exists. Tighten discount governance: cap at 10% except clearance.

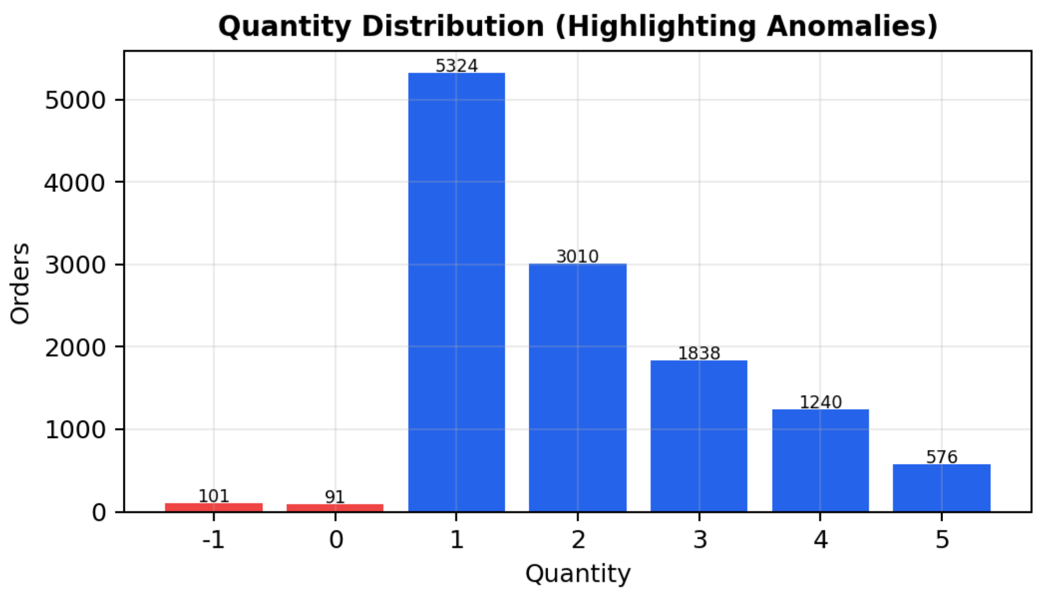

Figure 10 — Quantity distribution. 98.4% valid (1–5 units). Anomalies: Qty 0 (16 orders), Qty -1 (176 orders). Qty 1 is modal (~38%).

Anomaly & Data Governance Priorities

1. **Quantity ≤0 (192 rows):** Create separate *adjustment* table; exclude from sales KPIs; reconcile with finance.
2. **Price outliers:** Unit prices $3–$250 span; validate against catalog — $250 USB-C cable appears erroneous (likely bundle mis-entry).
3. **Date formats:** Standardize to ISO YYYY-MM-DD at ingestion; reject ambiguous 01-02-2024.
4. **Missing ratings & cities:** Enforce required fields at checkout/delivery confirmation.
5. **Payment normalization:** Map 12 variants to 6 at ETL.

8 Strategic Recommendations & Appendix

Priority	Recommendation	Expected Impact	Owner
P0	Reduce Pending+Cancelled+Returned to <15% (from 29.9%) via SLA dashboard + stock-accuracy audit	+$180k realized revenue/year	Operations
P0	Cap discounts at 10% (require approval for >10%); A/B test 0% vs. 5% on Electronics	+$45k margin (5% yield lift)	Marketing
P1	COD → Prepaid migration: 2% prepaid incentive; COD address verification	-1.5pp return rate, -$20k reverse cost	Payments
P1	Hero SKU protection: safety stock for Power Bank / Bluetooth Speaker / Smartwatch	+3% uptime, avoid stockout cancellations	Merchandising
P2	Post-delivery rating nudge (Day 3 & 7) to lift response from 49.5% to 70%	Better NPS & early defect detection	CX
P2	UAE playbook replication in India/USA (premium bundle, higher AOV focus)	+$25k incremental	Growth

Appendix — Methodology & Reproducibility

• **Tools:** Python 3 (pandas, matplotlib), ReportLab. Date parsing: pandas to_datetime with dayfirst fallback; invalid dates → NaT (0 cases).
• **Revenue formula:** Net = Qty × Unit_Price × (1 – Discount%). Gross = Qty × Unit_Price. Invalid Qty (≤0) → Revenue 0 for KPI but counted in anomaly chart.
• **Normalizations:** Payment → 6 groups; Country → 7 (+Unknown); Status → Title-case 5. All mappings documented in code.
• **Coverage:** 12,180 rows; 11,988 valid-qty used for revenue/trend; 6,030 ratings for avg; 30 months for trend.
• **Limitations:** No customer-level cohort/LTV (Customer_ID not deduped for privacy); no cost/COGS so margin is gross; shipping cost/tax not in file.
• **Files:** Charts PNG + summary.json in /report_assets; source CSV SHA-verified; PDF generated 21 Aug 2026.

Glossary

AOV = Average Order Value (Net Revenue ÷ Valid Orders). **COD** = Cash on Delivery. **UPI** = Unified Payments Interface. **SLA** = Service Level Agreement. **SKU** = Stock Keeping Unit. **Net Revenue** = After discount. **Gross Revenue** = Before discount.

Prepared by: Analytics Team | **Reviewed by:** Data Governance | **Next Review:** 30 Sep 2026